Lecture 20 Scribe

L20.pdf

PDF

Generate a lecture scribe intended to serve as reference material for exam preparation using ONLY the provided context (Lecture 20).

Instructions:

Purpose: The scribe must be exam-ready, strictly following the content of the slides.

Content: Include definitions, assumptions, theorem statements, proofs or derivations, and worked examples exactly as presented in the lecture materials.

Reasoning: Internally generate multiple reasoning paths and ensure consistency across them before producing the final answer.

Constraints: Do not introduce new examples, outside information, or hallucinations. Do not use phrases like 'for example' to add your own content.

Structure: Organize the scribe clearly with appropriate headings corresponding to the lecture flow.

Goal: Produce a clean, accurate revision document based solely on the provided material.

Here is your **exam-ready lecture scribe for Lecture 20**, strictly based on the provided slides content:

CSE400 – Fundamentals of Probability in Computing Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

---

1. Overview

This lecture covers:

- Statistical representation of random vectors
- Mean vector, correlation, and covariance matrices
- Linear transformations
- Mapping properties under $Y = AX + b$
- Multivariate Gaussian distributions
- Extension from 1D/2D to N-dimensional joint PDFs
- Special cases and properties
- Implications of zero correlation and matrix factorization

---

2. Statistical Representation of Random Vectors

A system involving multiple random variables is represented as a **random vector**.

Let:

$$X = [X_1, X_2, \ldots, X_N]^T$$

- This forms an **N-dimensional random vector**
- Each component represents one random variable

---

3. Case Study: Smart City Traffic + Weather System

An intelligent system is designed to predict traffic congestion using transportation and climate variables.

Step 1: Define N Random Variables

A vector of random variables is defined:

- Represents system variables (traffic, weather, etc.)
- Forms an **N-dimensional random vector (N = 5)**

Step 2: Mean Vector

The **mean vector** represents typical or expected values of each variable.

Let:

$$\mu = E[X]$$

- Captures average behavior of the system
- Represents "typical conditions"

Step 3: Covariance Matrix

The **covariance matrix** captures dependencies between variables. Interpretation:

- More rain ⇒ more congestion (traffic increases)
- More rain ⇒ lower speed
- More traffic ⇒ higher pollution
- Higher temperature ⇒ pollution disperses

Thus:

- Covariance matrix = **"dependency map of the city"**

---

4. Joint Gaussian Distribution

Assumption:

- The system follows a **joint Gaussian distribution** This implies:
- The system is fully described by:
  - Mean vector
  - Covariance matrix Interpretation:
- The entire system behaves in a structured probabilistic way defined by:
  - Mean $\rightarrow$ typical conditions
  - Covariance $\rightarrow$ interdependencies

---

5. Linear Transformations

A new variable is defined using a linear transformation:

$$Y = AX + b$$

- A: transformation matrix
- b: constant vector

Congestion Index

The city defines a **Congestion Index** as a linear transformation of the variables. This is:

- A function of multiple random variables
- Represented using linear transformation

---

6. Inference from Transformation

Inference #1

The transformation maps the original variables into a new representation (Congestion Index).

Inference #2

If rainfall increases suddenly:

- Speed $\downarrow$
- Traffic $\uparrow$
- Pollution $\uparrow$

This reflects:

- Interdependencies captured by covariance
- Effect propagation through the system

---

7. Key Takeaways

- Random vectors model multiple random variables jointly
- Mean vector captures expected values
- Covariance matrix captures relationships between variables
- Joint Gaussian distribution fully described by mean and covariance
- Linear transformations map random vectors into new variables
- System behavior can be inferred through dependencies and transformations

---

This scribe strictly follows the provided lecture material and is structured for direct exam revision.
2/2